

S&P 500 Quantitative Data Analysis — Master Glossary & Lexicon (Grand Lexique Bilingue)

Course: Quantitative Data Analysis (M1 – S7, Course code: 2627_ECO_2_EN_009)
Institution: EM Normandie Business School · Programme Grande École **Lecturer:** Dr. NGUYEN Anh-Tuan **Purpose:** A comprehensive, beginner-friendly bilingual guide (English concepts with detailed French explanations) covering all Financial, Data, Statistical, Econometric, ESG, and Jamovi terminology across all 4 sessions of the project.

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1. Data Fundamentals & Variable Typology

Unit of Observation (Unité d'observation)

- **Definition:** Exactly what a single row represents in your database.
- **In our project: 1 row = 1 publicly traded S&P 500 company.**
- **Why it matters (Slide 11):** If you cannot finish the sentence “*One row = one...*”, you do not understand your dataset. Having 1 row = 1 firm ensures independence of observations for ANOVA and regression.

Sample Size (N / Observation count)

- **Definition:** The total number of valid entities observed in the study.
- **In our project:** $N = 503$ companies (exceeds the course minimum requirement of $N \geq 100$).

Qualitative / Categorical Variable (Variable qualitative)

Variables that express names, categories, or labels rather than numeric quantities. You can count their frequency, but you **cannot** compute a mathematical average.

- **Nominal Variable (Variable nominale):**
 - Categories with **no natural order or ranking**.
 - *Examples:* Industry sector (Sector), country of listing, legal status.
 - *Course rule (Slide 29):* Must have between **2 and 5 categories**. Our project has exactly 5: *Tech & Comms, Healthcare, Finance, Industrials & Energy, Consumer*.

- *Golden Rule (Slide 19): Never compute a mean for a nominal variable!* Saying “Average sector = 2.4” is completely meaningless and results in lost marks.
- **Ordinal Variable (Variable ordinaire):**
 - Categories that possess a **clear, natural ranking or order**, but where the distance (gap) between categories is **not equal** or mathematically measurable.
 - *Examples:* Credit rating (AAA > AA > A), Customer satisfaction (1 star to 5 stars), Governance tier (*Low* < *Medium* < *High*), Size quartile ($Q1 < Q2 < Q3 < Q4$).
 - *Golden Rule (Slide 13):* The distance between A and B is not the same as between B and C. Central tendency is captured by the **mode** or the **median category**.

Quantitative / Numerical Variable (Variable quantitative)

Variables that represent measurable or countable numerical values where arithmetic operations (addition, subtraction, averaging) are valid.

- **Discrete Variable (Variable discrète):** Countable integer values with no decimals (e.g., Number of employees Headcount, number of factories, number of board members).
- **Continuous Variable (Variable continue):** Measured on an unbroken numerical continuum, allowing decimals (e.g., Profit_Margin, Total_Revenue_B, Market_Cap_B, Beta).
- **Scale Types: Interval vs. Ratio (Échelle d'intervalle vs. Échelle de rapport):**
 - *Interval scale:* Has an arbitrary zero point (e.g., Temperature in °C: 0°C does not mean “absence of temperature”, so 20°C is not “twice as hot” as 10°C).
 - *Ratio scale:* Has a **true, absolute zero** indicating complete absence of the quantity (e.g., Turnover = \$0 means zero sales; \$20M is truly twice as much as \$10M).
 - *In our project:* Total_Revenue_B, Market_Cap_B, and Profit_Margin are ratio-scale continuous variables, enabling full correlation and regression modeling.

Dataset Structure: Cross-Sectional vs. Time-Series vs. Panel

- **Cross-Sectional Dataset (Coupe transversale):** Multiple entities (503 firms) observed at a **single point in time** (e.g., FY2025). This is our project structure.
- **Time-Series (Série temporelle):** A single entity observed over multiple consecutive periods (e.g., Apple’s stock price daily from 2020 to 2026).
- **Panel Data (Données de panel):** Multiple entities observed across multiple time periods (1 row = 1 firm-year). *Caution:* Panel data introduces autocorrelation, which violates standard ANOVA and OLS regression assumptions taught in this course.

Outliers: Detection & Handling (Valeurs aberrantes / extrêmes)

Data points that deviate substantially from the overall pattern of the distribution. Outliers can distort arithmetic means, inflate standard deviations, and invalidate normality:

- **Tukey’s $1.5 \times IQR$ Rule (Règle des moustaches de Tukey):**
 - Any observation below the lower fence: $LF = Q_1 - 1.5 \times IQR$
 - Any observation above the upper fence: $UF = Q_3 + 1.5 \times IQR$
 - Points outside this range appear as individual dots outside the boxplot whiskers (e.g., NVIDIA or Apple’s extreme market cap; highly negative profit margins in turnaround firms).
- **Z-Score Method (Seuil en écarts-types):**
 - Standardized distance: $z_i = \frac{x_i - \bar{x}}{s}$. An observation with $|z_i| > 3.0$ is traditionally flagged as an extreme outlier in normally distributed data.

- **Management Strategy (Slide 21):** Never delete legitimate business outliers (like mega-cap firms) without justification! Instead, document them, apply non-parametric/robust tests (Median, IQR, Welch ANOVA), or use logarithmic transformations.

Missing Data: Mechanisms & Treatment (Données manquantes)

- **Missing Completely at Random (MCAR — Manquant complètement au hasard):** The probability of missingness is unrelated to any observed or unobserved variable (e.g., accidental scraping network drop on a random ticker).
 - **Missing at Random (MAR — Manquant au hasard conditionnel):** Missingness depends on other observed variables (e.g., smaller firms failing to disclose certain governance sub-scores).
 - **Missing Not at Random (MNAR — Manquant non au hasard):** Missingness depends on the unobserved value itself (e.g., firms with catastrophic ROE having negative equity, making ROE mathematically undefined).
 - **Treatment Strategies:**
 - **Listwise Deletion (Suppression par liste / Complete Case Analysis):** Dropping any row containing a missing value across the analyzed variables. Safe when MCAR and sample size is large ($N = 503$).
 - **Pairwise Deletion (Suppression par paire):** Used in bivariate correlation matrices (calculating r between variables X and Y using all rows where both X and Y are valid, preserving sample size per pair).
 - **Data Imputation (Imputation de données):** Replacing missing values with mean/median or predictive regression (avoided here to prevent artificial variance deflation).
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2. Financial & Corporate Accounting Terminology

S&P 500 Index (Standard & Poor's 500)

A market-capitalization-weighted equity index tracking the 500 largest public companies listed on stock exchanges in the United States (NYSE and NASDAQ), representing approximately 80% of total U.S. equity market capitalization.

Ticker Symbol (Symbole boursier)

A unique arrangement of 1 to 5 letters identifying a specific publicly traded security on an exchange (e.g., AAPL for Apple, MSFT for Microsoft, NVDA for Nvidia, FI for Fiserv). Serves as the primary ID variable in our dataset.

Market Capitalization (Market_Cap_B — Capitalisation boursière)

$$\text{Market Cap} = \text{Current Share Price} \times \text{Total Shares Outstanding}$$

The total equity market value of a public company, expressed in **Billions of USD (\$B)** in our dataset.

- **Market Cap Quartiles (Market_Cap_Quartile):** The S&P 500 divided into 4 equal tiers of firm size:
 - **Q1:** Smallest quartile in the index (\$6.5B to \$22.2B)
 - **Q2:** Mid-sized large-cap (\$22.2B to \$44.5B)
 - **Q3:** Upper large-cap (\$44.5B to \$94.0B)

- **Q4:** Mega-cap giants (\$94.0B to \$5,367.1B, including Apple, Microsoft, Nvidia)

Total Revenue (Total_Revenue_B — Chiffre d'affaires / Turnover)

The total gross amount of income generated by the sale of goods or services related to the company's primary operations before deducting any expenses, taxes, or depreciation. Reported in **Billions of USD (\$B)**.

Fiscal Year (FY — Exercice fiscal) vs. Calendar Year (Année calendaire)

- *Calendar Year:* Strictly January 1 to December 31.
- *Fiscal Year (FY2025):* The official 12-month accounting period used by a corporation to prepare audited financial reports (SEC Form 10-K).
- *Why it matters:* Many US companies do not close their books on December 31. Apple closes in September; Nvidia and Walmart close in January; Microsoft closes in June. Using **FY2025** ensures we analyze 12 full months of audited operating figures for every single corporation.

Trailing Twelve Months (TTM — Douze derniers mois glissants)

A financial reporting measurement window that calculates performance over the most recent 12-month period (the last 4 quarters) regardless of the fiscal year-end date.

Net Profit Margin (Profit_Margin — Marge bénéficiaire nette)

$$\text{Profit Margin} = \frac{\text{Net Income}}{\text{Total Revenue}}$$

The percentage of revenue that remains as bottom-line profit after paying all operating expenses, interest, and taxes. Expressed as a decimal in our dataset (e.g., 0.15 = 15%).

- *Advantage in Data Analysis:* Revenue is positive for all operating firms, meaning Profit_Margin is 100% defined and available for all 503 firms without accounting distortion.

Return on Equity (ROE — Rentabilité des capitaux propres)

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

Measures how efficiently a company's management uses shareholders' capital to generate net profits. Expressed as a decimal (e.g., 0.20 = 20%).

Negative Shareholders' Equity & Missing ROE (Capitaux propres négatifs)

- *The Phenomenon:* In our dataset, 32 prominent corporations (AbbVie, Altria, AutoZone, Booking Holdings, Domino's Pizza, etc.) show missing/NaN values for ROE.
- *The Financial Reason:* These mature, highly cash-generative firms engaged in aggressive, debt-funded **share buybacks (rachats d'actions)** over 10+ years. Buying back stock reduces the accounting book equity on the balance sheet. When cumulative buybacks exceed retained earnings, book equity becomes negative.
- *Econometric Implication:* Net income divided by negative equity produces a mathematically meaningless or negative number. Financial databases (Yahoo Finance, Bloomberg) set ROE to NaN. Rather than deleting these premier companies (which would introduce survival and industry bias), we document this in the *Data Cleaning Log* and utilize Profit_Margin as our universal profitability indicator.

Market Beta (Beta — Volatilité systématique)

A quantitative measure of a stock's volatility (systematic market risk) in comparison to the overall market (S&P 500 index), calculated from a **5-year window of monthly returns**:

- $\beta = 1.0$: Stock moves in tandem with the market.
- $\beta > 1.0$: Aggressive / high-volatility stock (e.g., high-growth tech).
- $\beta < 1.0$: Defensive / low-volatility stock (e.g., consumer staples, utilities).
- $\beta < 0$: Inverted movement (extremely rare in equities).

Form 10-K (Rapport annuel SEC)

The comprehensive annual regulatory filing required by the U.S. Securities and Exchange Commission (SEC) providing a detailed, audited breakdown of a public company's financial performance.

GICS (Global Industry Classification Standard — Classification sectorielle)

An industry taxonomy developed in 1999 by MSCI and S&P Dow Jones Indices. Classifies all public companies into **11 primary sectors**: *Information Technology, Communication Services, Financials, Health Care, Consumer Discretionary, Consumer Staples, Industrials, Energy, Utilities, Real Estate, Materials*.

- **Course Mapping (Slide 29)**: The course strictly limits the nominal grouping factor to 2–5 categories. We consolidated the 11 raw GICS sectors into **5 economically coherent groups**: *Tech & Comms, Healthcare, Finance, Industrials & Energy, Consumer*.

The Fundamental Accounting Equation (Équation fondamentale du bilan)

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$$

$$\text{Actif} = \text{Dettes} + \text{Capitaux Propres}$$

A company's economic resources (Assets) are financed either by borrowing from creditors (Liabilities) or by capital provided and accumulated by owners (Equity). When liabilities exceed assets (often due to massive debt-funded share repurchases), accounting equity turns negative.

The P&L Income Statement Waterfall (Cascade du compte de résultat)

The sequential deduction of operating and non-operating costs to arrive at net bottom-line earnings:

1. **Gross Revenue (Chiffre d'affaires)**: Total operational sales inflow (Total_Revenue_B).
2. **Gross Profit (Marge brute)**: Revenue minus Cost of Goods Sold (COGS).
3. **EBITDA (Excédent Brut d'Exploitation)**: Earnings Before Interest, Taxes, Depreciation, and Amortization. A pure measure of operational cash generation.
4. **EBIT / Operating Income (Résultat d'exploitation)**: Operational profit after depreciation of physical plant and equipment.
5. **EBT (Résultat avant impôts)**: Operating income minus interest on debt.
6. **Net Income (Résultat net)**: Final accounting earnings distributed to shareholders after corporate income tax. Divided by revenue, this yields our key outcome variable: Profit_Margin.

Systematic Risk vs. Idiosyncratic Risk (CAPM / MEDAF)

Under the Capital Asset Pricing Model (CAPM):

- **Systematic Market Risk (Risque systématique — β)**: Macroeconomic risks affecting the entire market (interest rates, inflation, recessions, geopolitical crises). Cannot be eliminated by diversification.
- **Idiosyncratic / Unsystematic Risk (Risque spécifique / idiosyncratique)**: Firm-specific risks (e.g., product failure, executive fraud, board scandal). Completely diversifiable in a large portfolio.
- **Our Empirical Finding**: Poor corporate governance is often assumed to be purely idiosyncratic, but our Session 2 correlation proves that weak governance significantly inflates **systematic market sensitivity** ($r = +0.268, p < .001^{***}$)!

Earnings Per Share (EPS / BPA) & Financial Engineering

$$\text{EPS} = \frac{\text{Net Income}}{\text{Total Shares Outstanding}}$$

Because executive bonuses and stock option vestings are heavily tied to EPS targets, corporate executives often use low-interest corporate debt to buy back company shares from the open market. This reduces the denominator (Shares Outstanding), artificially boosting EPS even when top-line business growth is stagnant.

Weighted Average Cost of Capital (WACC / CMPC)

The average after-tax rate of return a company must pay to all its security holders (debt-holders and equity-holders) to finance its assets. Poor corporate governance increases perceived risk among lenders and investors, resulting in a **governance risk premium** that elevates the firm's WACC and reduces valuation.

3. ESG & Corporate Governance (ISS Framework)

ESG (Environmental, Social, and Governance)

A tri-pillar framework used by institutional investors to evaluate corporate sustainability, societal impact, and ethical leadership alongside traditional financial statements.

ISS (Institutional Shareholder Services)

The global leader in proxy voting advisory and corporate governance research. ISS evaluates publicly listed firms and assigns quantitative risk ratings based on regulatory disclosures.

ISS Governance QualityScore (Overall_Governance_Risk)

A composite decile score measuring corporate governance risk across 4 core pillars on a **1 to 10 scale**:

- **Audit Risk (Audit_Risk)**: Financial integrity, accounting oversight, auditor independence, and financial restatements.
- **Board Structure Risk (Board_Risk)**: Independence of directors, board diversity, committee structures, and separation of CEO and Chairman roles.

- **Compensation Risk (Compensation_Risk):** Alignment between executive pay and shareholder returns, clawback policies, and golden parachutes.
- **Shareholder Rights Risk (Shareholder_Rights_Risk):** Voting rights, poison pills, classified boards, and takeover defense mechanisms.

[!CAUTION] **Crucial Inversion:** In the ISS methodology, **1 represents the lowest risk (best governance practices)**, while **10 represents the highest risk (poorest governance practices)**! A negative correlation between Governance Risk and Profit Margin therefore implies that *better governance (lower risk score) is associated with higher profitability*!

Governance Risk Tier (Governance_Risk_Level — Variable ordinale)

To enable categorical contingency tables (Chi-Square) and group tests, the 1–10 continuous score is classified into three ordered tiers:

- **Low Risk** (Scores 1 to 3): Premier governance compliance ($n = 150$, 30.2%).
- **Medium Risk** (Scores 4 to 6): Standard market governance ($n = 149$, 30.0%).
- **High Risk** (Scores 7 to 10): Elevated governance friction ($n = 197$, 39.7%, Modal tier).

Agency Theory & Corporate Governance (Théorie de l'agence)

The foundational financial theory of corporate governance (Jensen & Meckling, 1976; Fama & Jensen, 1983):

- **The Core Conflict:** The **separation of ownership** (principals = shareholders who provide capital) from **operational control** (agents = managers and CEOs who run the day-to-day business).
- **Agency Costs (Coûts d'agence):** Self-interested managers may maximize their own private benefits (excessive pay, corporate jets, empire building through unprofitable mergers) rather than maximizing long-term shareholder value.
- **The Role of Governance:** Board oversight, independent auditing, executive pay alignment, and shareholder voting rights exist precisely to minimize agency friction and ensure managers work in the interest of owners.

Key Governance Mechanisms Monitored by ISS

- **CEO Duality (Cumul des mandats de PDG et Président du Conseil):** When the Chief Executive Officer also serves as Chairman of the Board of Directors. ISS considers this a major conflict of interest because the board's primary role is to evaluate and supervise the CEO. Having an **Independent Lead Director** mitigates this risk.
- **Say-on-Pay (Vote consultatif sur la rémunération):** A non-binding shareholder vote mandated under the Dodd-Frank Act allowing investors to approve or reject executive compensation packages. Repeated low approval (< 70%) triggers elevated ISS Compensation Risk.
- **Clawback Provision (Clause de restitution des bonus):** A contractual clause enabling the board to claw back previously awarded executive bonuses in the event of financial restatements, fraud, or material misconduct.
- **Poison Pill / Shareholder Rights Plan (Pilule empoisonnée):** A defense mechanism allowing existing shareholders to purchase newly issued shares at a steep discount during

an uninvited hostile takeover bid, severely diluting the hostile acquirer. ISS heavily penalizes companies adopting poison pills without shareholder approval.

- **Classified / Staggered Board (Conseil d'administration échelonné)**: A board structure where only a fraction (typically one-third) of directors stand for election each year. Prevents an activist shareholder or acquirer from replacing the entire board in a single proxy contest.

Proxy Advisory & The “Big Three” Asset Managers

- **Proxy Advisors (Agences de conseil en vote)**: Firms like ISS and Glass Lewis that research shareholder proposals, analyse corporate filings, and issue vote recommendations for annual general meetings (AGMs).
 - **The Big Three (BlackRock, Vanguard, State Street)**: The largest passive index asset managers in the world, holding on average over 20% of the voting shares of S&P 500 corporations. Because they manage passive index funds and cannot simply “sell” the stock of a poorly run company, they rely heavily on **ISS governance scores and proxy voting** to force corporate governance reform.
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4. Session 1: Univariate Statistics & Shape Parameters

Univariate Analysis (Analyse univariée)

The statistical examination of **one single variable in isolation**, without analyzing causes, relationships, or predictive influences (Slide 19).

Central Tendency: The Three Centres (Tendance centrale)

1. **Arithmetic Mean ($\bar{x}$ – Moyenne)**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mathematical average. Takes every data point into account. **Weakness (Slide 21)**: Completely distorted and destroyed by extreme outliers and skewness.

2. **Median (Mdn – Médiane)**: The exact middle value (50th percentile) dividing the ordered dataset into two equal halves. **Strength**: Robust and immune to extreme outliers.
3. **Mode (Mode)**: The most frequently occurring value or category in a distribution. **The only valid measure of central tendency for nominal data** (Slide 20).

Dispersion & Variability (Mesures de dispersion)

- **Standard Deviation (s – Écart-type)**:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

The average distance of an observation from the mean. Tells you whether the mean accurately represents the typical case (Slide 22).

- **Variance (s^2)**: The square of the standard deviation.
- **Range (Étendue)**: The absolute distance between maximum and minimum (Max – Min).
- **Interquartile Range (IQR – Écart interquartile)**:

$$IQR = Q_3 - Q_1$$

The range covering the central 50% of observations. Highly robust against outliers.

Distribution Shape: Skewness & Kurtosis (Forme de distribution)

- **Skewness (g_1 – Asymétrie):**

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Measures the lack of symmetry in a distribution.

- $g_1 \approx 0$: Symmetric distribution (Mean $\approx$ Median $\approx$ Mode).
- $g_1 > 0$ (**Positive / Right Skew**): Mean $>$ Median $>$ Mode. A long tail extending to the right (typical of income, revenue, market cap).
- $g_1 < 0$ (**Negative / Left Skew**): Mean $<$ Median $<$ Mode. A long tail extending to the left (typical of profit margins with distressed outliers).

- **Kurtosis (g_2 – Aplatissement / Acuité):**

$$g_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

Measures the “tailedness” and peakedness of the distribution relative to a normal curve.

- $g_2 = 0$ (Mesokurtic): Standard bell-shaped normal curve.
- $g_2 > 0$ (Leptokurtic): Sharply peaked with heavy, fat tails (higher probability of extreme outliers).
- $g_2 < 0$ (Platykurtic): Flatter peak with thin tails.
- **The Normality Rule of Thumb (Slide 23):** Both skewness and kurtosis falling within $[-1.0, +1.0]$ is consistent with normality.

Normality Testing: The Three Converging Pieces of Evidence (Slide 24)

Never rely on a single metric; evaluate all three simultaneously:

1. **Compare the Three Centres:** If Mean $\approx$ Median $\approx$ Mode, the distribution is roughly symmetric.
2. **Evaluate Skewness and Kurtosis:** Verify whether both reside inside the $[-1.0, +1.0]$ window.
3. **Visual Inspection & Formal Shapiro-Wilk Test:**
 - *Visual Inspection:* Inspect Histogram, Density curve, and Q-Q Plot.
 - *Shapiro-Wilk Test:* Formal statistical test with null hypothesis H_0 : “The data are normally distributed”.
 - **The Inverted Logic (Slide 24):** Unlike almost every other test in the course, $p > 0.05$ means you **KEEP normality**. If $p < 0.05$, you **REJECT normality**.

5. Session 2: Bivariate Associations, Chi-Square & Correlations

Bivariate Analysis (Analyse bivariable)

The statistical examination of the relationship or association between **two variables simultaneously** (Slide 17 & Session 2).

The Four-Step Protocol (Slide 9 & 41)

The mandatory answer template for every analytical question:

1. **IDENTIFY:** Which statistical test to execute? Fully justified by the variable scales.
2. **HYPOTHESE:** Formulate H_0 and H_1 in clear, non-technical business language.
3. **DECIDE:** Is $p < 0.05$? If yes, what is the magnitude/strength of the effect?
4. **RECOMMEND:** Actionable executive guidance answering: “What does the manager do on Monday morning?”

Hypotheses: Null (H_0) vs. Alternative (H_1)

- **Null Hypothesis (H_0):** The hypothesis of “no difference”, “no relationship”, or “status quo” (e.g., “Governance risk has no relationship with sector”).
- **Alternative Hypothesis (H_1):** The research hypothesis proposing an active relationship or group difference (e.g., “Governance risk varies significantly across sectors”).

Statistical Significance (p -value) vs. Effect Size (Slide 42)

- **p -value (Significance):** The probability of observing results at least as extreme as the sample data assuming H_0 is true. If $p < 0.05$, we reject H_0 and conclude that an effect exists.
- **Reporting Standard (Slide 42):** Never write $p = 0.000$! Software printing .000 means the value is below three decimal places. Report strictly as $p < .001$.
- **Effect Size (Strength):** While p -value tells you *whether* an effect exists, effect size tells you *how strong* it is (whether it practically matters to a manager).

Decision Errors: Type I (α) vs. Type II (β) & Statistical Power

In hypothesis testing, two types of erroneous conclusions are possible:

Reality	Decision	Fail to Reject H_0 (Accept Status Quo)	Reject H_0 (Conclude Effect Exists)
H_0 is True (No real effect)	Correct Decision	Decision ($1 - \alpha = 95\%$)	Type I Error ($\alpha = 5\%$) (False Positive)
H_0 is False (Real effect exists)	Type II Error (β) (False Negative / Missed Discovery)		Correct Decision: Statistical Power ($1 - \beta \geq 80\%$)

- **Type I Error ($\alpha = 0.05$ – Faux positif):** Claiming a governance relationship exists when it is purely due to random chance. The significance threshold α fixes this maximum acceptable risk at 5%.
- **Type II Error (β – Faux négatif):** Missing a genuine governance impact because the sample is too small or noisy.
- **Statistical Power ($1 - \beta$ – Puissance statistique):** The probability of successfully detecting a genuine effect. By having $N = 503$ (well above the course minimum of 100), our study achieves high statistical power ($> 95\%$), minimizing Type II errors.

95% Confidence Interval (Intervalle de confiance à 95% – CI)

$$CI_{95\%} = \text{Point Estimate} \pm t_{\text{crit}} \times \text{Standard Error}$$

A range of values calculated from the sample data that has a 95% probability of containing the true population parameter:

- **Interpretation Rule:** If the 95% Confidence Interval for a difference or correlation contains zero (e.g., $[-0.05, +0.12]$), the effect is **not statistically significant** at $\alpha = 0.05$. If zero is excluded (e.g., Pearson r for Governance Risk and Beta: $[+0.18, +0.35]$), the relationship is statistically significant.
- **Margin of Error (Marge d'erreur):** Half the width of the confidence interval ($t_{\text{crit}} \times SE$), quantifying sampling uncertainty.

Chi-Square Test of Independence (χ^2 – Nominal $\times$ Ordinal)

Tests whether two qualitative categorical variables are independent or statistically associated.

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where O_{ij} is the observed cell count, and $E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{N}$ is the expected count under independence. Degrees of freedom: $df = (r - 1)(c - 1)$.

- **Cramér's V (Effect Size for Chi-Square):**

$$V = \sqrt{\frac{\chi^2}{N \min(r - 1, c - 1)}}$$

Measures association strength from 0 (no relationship) to 1 (perfect association):

- $V < 0.10$: Negligible
 - $0.10 \leq V < 0.30$: Weak to moderate association (Our result: $V = 0.122$)
 - $V \geq 0.30$: Strong association
- **Standardized Residuals (Résidus standardisés):**

$$\text{Residual}_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

Identifies which specific cells drive the overall Chi-Square result. A residual $> +2.0$ indicates significant over-representation; < -2.0 indicates significant under-representation. (In our data, Tech & Comms had $+1.75$ in High Risk; Industrials had -1.97).

Pearson Correlation Coefficient (r – Continuous $\times$ Continuous)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Measures the linear strength and direction of the association between two continuous variables:

- Ranges from -1.0 (perfect inverse link) to $+1.0$ (perfect direct link).
- **Fisher z -Transformation & 95% Confidence Interval:** Constructs exact lower and upper confidence bounds:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right), \quad SE_z = \frac{1}{\sqrt{n-3}}$$

Spearman's Rank Correlation (ρ – Non-Paramétrique)

Calculates Pearson's correlation on the **ranks** of data rather than raw values. Robust to non-normality and monotonic non-linear patterns.

6. Session 3: Group Comparisons, Welch's ANOVA & Games-Howell

One-Way Analysis of Variance (ANOVA – Qualitative 3+ groups $\times$ Quantitative)

Tests whether the population means of a continuous outcome variable differ significantly across 3 or more categorical groups.

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)}$$

Levene's Test of Homogeneity of Variances (Test de Levene)

Tests whether variances are equal across groups ($H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$).

- If $p > 0.05$: Variances are equal (homoscedasticity confirmed; standard Fisher ANOVA is valid).
- If $p < 0.05$: Variances are significantly unequal (**heteroscedasticity**; standard Fisher ANOVA is invalid and Welch's ANOVA must be used!).
- *In our project*: Levene's test yielded $F = 5.39, p < .001$ for Profit Margin across sectors.

Welch's Robust ANOVA (F_{Welch})

An adjusted ANOVA formulation that weights each group's variance by its sample size ($w_i = n_i/s_i^2$). Valid even when group variances and sample sizes are severely unequal.

Effect Size: Eta-Squared (η^2) & Omega-Squared (ω^2)

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

The percentage of variance in the dependent variable accounted for by group membership:

- $\eta^2 \approx 0.01$: Small effect
- $\eta^2 \approx 0.06$: Medium effect
- $\eta^2 \geq 0.14$: Large effect
- *In our project*: $\eta^2 = 0.0405$ (4.05% of profit margin variance is explained by sector).

Kruskal-Wallis Non-Parametric Test (H)

The non-parametric rank-based equivalent of One-Way ANOVA. Used when both normality and variance homogeneity are severely violated.

Post-Hoc Pairwise Comparisons: Tukey HSD vs. Games-Howell

When ANOVA indicates that at least one group differs ($p < 0.05$), post-hoc tests identify *which specific pairs* differ:

- **Tukey's HSD (Honestly Significant Difference):** Assumes equal variances across all groups.
- **Games-Howell Post-Hoc Test:**

$$t = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\frac{s_i^2}{n_i} + \frac{s_j^2}{n_j}}}$$

Adjusts degrees of freedom via the Welch-Satterthwaite equation and applies the Studentized Range distribution (q). **Required when Levene's test fails!**

7. Session 4: Multiple Linear Regression & Econometric Diagnostics

Multiple Linear Regression (Régression linéaire multiple MCO / OLS)

Models the expected value of a continuous dependent variable (Y) as a linear combination of explanatory variables ($X_1, X_2, \dots, X_k$):

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i$$

- β_0 (**Intercept / Constante**): The predicted baseline value of Y when all predictors are equal to zero.
- β_k (**Unstandardized Regression Coefficient**): The expected change in Y for a 1-unit increase in X_k , holding all other variables constant (*ceteris paribus*).
- **Standardized Beta (β^*)**:

$$\beta_k^* = \beta_k \times \frac{s_{X_k}}{s_Y}$$

Allows direct comparison of predictor importance on a common standard-deviation scale.

Coefficient of Determination (R^2 and Adjusted R^2)

- R^2 : The proportion of variance in Y explained by the regression model:

$$R^2 = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}}$$

- **Adjusted R^2** : Corrects R^2 for the number of predictors (k) and sample size (N):

$$R_{\text{adj}}^2 = 1 - \left[\frac{(1 - R^2)(N - 1)}{N - k - 1} \right]$$

Logarithmic Transformations ($\ln(X)$)

Applying natural logarithms to positively skewed financial metrics (Market Capitalization and Revenue).

- *Econometric Rationale*: Compresses extreme mega-cap outliers (NVIDIA, Apple) and transforms multiplicative scale dynamics into linear additive relationships.

Econometric Assumptions & Diagnostic Suite (Slide 133–137)

1. **Linearity:** Evaluated via the Residuals vs. Fitted plot (should show a flat horizontal band with no curvature).
2. **Normality of Residuals:** Model errors (ϵ) must be normally distributed (Q-Q plot of residuals).
3. **Homoscedasticity (Equal Error Variance):**
 - **Breusch-Pagan Test:** Regresses squared residuals on explanatory variables.
 - If $p < 0.05$: Heteroscedasticity is present (White's robust standard errors HC3 are required).
4. **Absence of Multicollinearity (VIF & Tolerance):**
 - **Variance Inflation Factor (VIF):**

$$VIF_j = \frac{1}{1 - R_j^2}$$

Measures how much the variance of an estimated regression coefficient increases due to collinearity with other predictors:

- $VIF < 5.0$: Safe / Low collinearity (Our maximum VIF was 1.89).
- $VIF > 10.0$: Severe multicollinearity (distorts t -statistics and inflates standard errors).
- **Tolerance:** The inverse of VIF ($Tol = 1/VIF$). Tolerance < 0.20 signals high collinearity risk.

Dummy Variables & Reference Category (Variables indicatrices / muettes)

To include a qualitative nominal variable (like Sector with 5 groups) in an OLS regression, it must be converted into $k - 1 = 4$ **binary dummy variables** taking values 0 or 1:

- **Reference Group (Catégorie de référence):** *Tech & Comms* is omitted from the equation to serve as the baseline ($X_{\text{all_dummies}} = 0$).
- **Interpretation of Dummy Coefficients (β_{Sector}):** The coefficient represents the expected difference in profit margin between that sector and the reference sector (Tech & Comms), holding all financial and governance variables constant.
- **Dummy Variable Trap (Piège de la colinéarité parfaite):** If all 5 dummy variables were included alongside the constant (β_0), the sum of dummies would equal 1, causing perfect multicollinearity ($VIF = \infty$) and rendering matrix inversion mathematically impossible.

Overall Model Utility: The Regression F -Test

Tests the global null hypothesis that *none* of the explanatory variables predict Y ($H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$):

$$F = \frac{MS_{\text{model}}}{MS_{\text{residual}}} = \frac{R^2/k}{(1 - R^2)/(N - k - 1)}$$

If $p_F < 0.05$, we reject H_0 and conclude that the model as a whole has genuine explanatory power beyond pure chance. (In our Model 3: $F(8, 485) = 11.23, p < .001^{***}$).

Influential Observations: Leverage (h_{ii}) & Cook's Distance (D_i)

Not all outliers exert the same distortive pull on regression slopes:

- **Leverage (h_{ii} – Valeur levier):** Measures how far an observation's predictor values (X) are from the center of the predictor space. High leverage points (e.g., massive revenue conglomerates) have high potential to swing the regression line. Warning threshold: $h_{ii} > 2(k+1)/N$.
- **Studentized Residual (r_i):** The residual divided by its estimated standard deviation, flagging outliers in the Y -dimension ($|r_i| > 3.0$).
- **Cook's Distance (D_i – Distance de Cook):**

$$D_i = \frac{\sum (\hat{y}_j - \hat{y}_{j(i)})^2}{(k+1)s^2}$$

Measures the aggregate shift in all model predictions when observation i is excluded. Any observation with $D_i > 1.0$ (or $D_i > 4/N$) is an **influential observation** requiring close inspection. (Panel D of Figure 9 visualizes studentized residuals against leverage).

Autocorrelation of Residuals & Durbin-Watson Test (d)

Assumption that error terms are independent ($\text{Cov}(\epsilon_i, \epsilon_j) = 0$):

- **Durbin-Watson Statistic (d):**

$$d = \frac{\sum_{i=2}^N (e_i - e_{i-1})^2}{\sum_{i=1}^N e_i^2}$$

Ranges from 0 to 4:

- $d \approx 2.0$: No autocorrelation (residuals are completely independent).
- $d < 1.5$: Positive autocorrelation (common in time-series data).
- $d > 2.5$: Negative autocorrelation.
- **In our Cross-Sectional Data:** Because 1 row = 1 independent firm, autocorrelation is naturally absent ($d \approx 1.98$).

8. Jamovi Software Operations & Costly Student Traps

What is Jamovi?

A free, open-source, point-and-click statistical software package built on top of the R statistical programming language, providing SPSS-style output tables and real-time reactive calculations (Slide 37).

Jamovi Measure Types (Slide 38)

- **ID:** Column used solely to identify entities (e.g., Ticker). Jamovi never computes statistics on ID columns.
- **Nominal:** Unordered qualitative categories (icon: three colored circles).
- **Ordinal:** Ranked qualitative categories (icon: small step chart).
- **Continuous:** Numeric measurements with decimals (icon: small ruler).

The .omv File Format

The proprietary Jamovi file format that bundles the spreadsheet dataset, variable configurations, and all statistical output tables and graphs together into a single file. **This is the primary file submitted for lab coursework** (Slide 37).

Top 6 Errors That Cost Marks Every Year (Slide 42)

1. **Reporting a mean for a nominal variable:** e.g., writing “*average sector* = 2.4”.
2. **Setting the wrong measure type in Jamovi:** e.g., setting Sector as Continuous or Profit_Margin as Nominal, causing Jamovi to hide the required statistical tests.
3. **Reading “Sig. = 0.000” as $p = 0$:** Always format as $p < .001$.
4. **Confusing significance with strength:** A test can be statistically significant ($p < 0.05$) because sample size N is large, but have a tiny, negligible effect size ($r = 0.08$).
5. **Pasting output tables without interpretation:** Half the marks are awarded for explaining what the numbers mean in plain English.
6. **Stopping at the statistic without recommending action:** Always conclude with what the executive or portfolio manager does on Monday morning.